WORKED SOLUTION

STEP III 2005, Question 12

Probability and Statistics • Combining Unbiased Estimators

STEP III 2005, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q12

The key idea

Everything rests on two facts about linear combinations of independent random variables: expectations add, and variances add with the squares of the coefficients. Requiring an expectation to come out right *whatever* the true lap times means matching coefficients, which turns each condition into an equation for the constants. Then X is a sensible estimate of the race time, and Y is a way of estimating the unknown σ from the disagreement between the two methods of timing.

Solution

Let the true lap times be $\mu_1, \ldots, \mu_4$ and the true race time $\mu = \mu_1 + \mu_2 + \mu_3 + \mu_4$. Write

$$S = T_1 + T_2 + T_3 + T_4.$$

Then $\mathrm{E}(S) = \mu_1 + \cdots + \mu_4 = \mu$ and, since $T_1, \ldots, T_4$ are independent (the timers are independent), $\mathrm{Var}(S) = 4\sigma^2$. Also $\mathrm{E}(T) = \mu$, $\mathrm{Var}(T) = \sigma^2$, and T is independent of S, because S depends only on the other four timers.

Finding X

For $X = aT + bS$,

$$\mathrm{E}(X) = a\mu + b\mu = (a+b)\mu.$$

This must equal μ whatever the true lap times, so in particular for some $\mu \neq 0$; hence

$$a + b = 1.$$

By independence of T and S,

$$\mathrm{Var}(X) = a^2 \mathrm{Var}(T) + b^2 \mathrm{Var}(S) = \sigma^2(a^2 + 4b^2) = \sigma^2(a^2 + 4(1-a)^2).$$

The quadratic $a^2 + 4(1-a)^2 = 5a^2 - 8a + 4$ has positive leading coefficient, so it is minimised where its derivative $10a - 8$ vanishes, at $a = \frac{4}{5}$. (Completing the square:

$5a^2 - 8a + 4 = 5(a - \frac{4}{5})^2 + \frac{4}{5}$.) Hence $a = \frac{4}{5}$, $b = \frac{1}{5}$, and

$$X = \tfrac{4}{5}T + \tfrac{1}{5}(T_1 + T_2 + T_3 + T_4), \qquad \mathrm{Var}(X) = \tfrac{4}{5}\sigma^2.$$

This makes sense: T is four times as reliable as S (variance σ^2 against $4\sigma^2$), so it gets four times the weight.

Finding Y

For $Y = cT + dS$ we use $\mathrm{E}(Y^2) = \mathrm{Var}(Y) + (\mathrm{E}(Y))^2$:

$$\mathrm{E}(Y^2) = \sigma^2(c^2 + 4d^2) + (c + d)^2\mu^2.$$

Ask yourself How can this equal σ^2 whatever the true lap times are?

The constants c, d may not depend on μ, but μ can be any positive number. The expression is $\sigma^2(c^2 + 4d^2)$ plus a multiple of μ^2, so for it to stay equal to σ^2 as μ varies the coefficient of μ^2 must vanish: $(c + d)^2 = 0$, so $d = -c$. Then we need $\sigma^2(c^2 + 4c^2) = \sigma^2$, that is, $c^2 = \frac{1}{5}$. Taking $c = 1/\sqrt{5}$,

$$Y = \frac{T - (T_1 + T_2 + T_3 + T_4)}{\sqrt{5}}$$

(the other sign gives $-Y$, which works equally well since only Y^2 matters). So Y measures the disagreement between the two timings; its expectation is 0, and its square is an unbiased estimate of σ^2.

The particular race

With $T = 220$ and $S = 220.5$,

$$X = \tfrac{4}{5}(220) + \tfrac{1}{5}(220.5) = 176 + 44.1 = 220.1,$$
$$Y = \frac{220 - 220.5}{\sqrt{5}} = -\frac{0.5}{\sqrt{5}}, \qquad Y^2 = \frac{0.25}{5} = 0.05.$$

So we estimate $\sigma^2 \approx 0.05$, and hence

$$\mathrm{Var}(X) = \tfrac{4}{5}\sigma^2 \approx \tfrac{4}{5}(0.05) = 0.04, \qquad \text{standard deviation of } X \approx 0.2 \text{ s}.$$

Our estimate of the true race time is therefore 220.1 ± 0.2 seconds: the true time lies roughly in the interval

$$[219.9,\ 220.3] \text{ seconds},$$

taking one standard deviation either side. If we wanted more confidence we could go two standard deviations either side, giving $[219.7,\ 220.5]$ seconds; for normally distributed errors this would be about a 95% interval. Bear in mind that σ^2 has been estimated from a single value of Y^2, so the width of the interval is itself only a rough estimate.

Checks

- With $a = \frac{4}{5}$, $\mathrm{Var}(X) = \frac{4}{5}\sigma^2$ is smaller than both $\mathrm{Var}(T) = \sigma^2$ and $\mathrm{Var}(S) = 4\sigma^2$, as combining two independent measurements should achieve.
- $X = 220.1$ lies between the two measurements, much closer to $T = 220$, the more reliable one.
- A simulation with 200 000 races ($\sigma = 0.7$) gave a mean of X within 0.001 of the true time, $\mathrm{Var}(X) \approx 0.393$ against $\frac{4}{5}(0.49) = 0.392$, and a mean of Y^2 of 0.491 against $\sigma^2 = 0.49$.

Common slips

- Taking $\mathrm{Var}(S) = \sigma^2$ or $16\sigma^2$: it is $4\sigma^2$, the sum of four independent variances.
- Forgetting the $(\mathrm{E}\,Y)^2$ term in $\mathrm{E}(Y^2)$, which is what forces $d = -c$.
- Using $Y^2 = 0.05$ as the variance of X, rather than scaling by $\frac{4}{5}$.

Beyond STEP

X is the minimum-variance unbiased linear combination of two estimates: the general rule is to weight each independent estimate by the reciprocal of its variance, which is the basis of inverse-variance weighting in meta-analysis and of the Gauss–Markov theorem for least squares. Y^2 is an unbiased estimator of σ^2, like the sample variance with its divisor $n - 1$. Grimmett and Welsh, *Probability: An Introduction*, covers expectation and variance of sums carefully, and a first statistics course takes up estimators and confidence intervals properly.

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